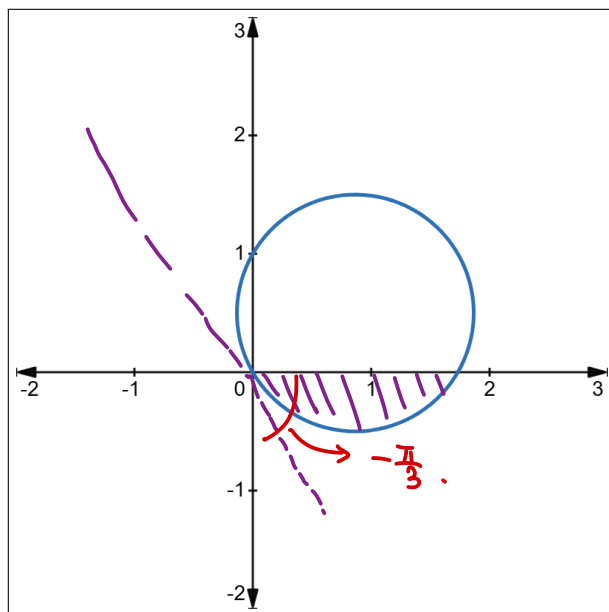


B5 past final worksheet

1. (Fall 2024, #9) Find the area of the segment of the circle with polar equation

$$r(\theta) = \sqrt{3} \cos \theta + \sin \theta; \quad \theta \in [-\pi/3, 2\pi/3]$$

below the polar axis (in the figure below, the polar axis coincides with the positive x -semiaxis and the pole with the Cartesian origin).



$$A = \int \frac{1}{2} r(\theta)^2 d\theta.$$

$$= \int_{-\pi/3}^0 \frac{1}{2} (\sqrt{3} \cos \theta + \sin \theta)^2 d\theta.$$

$$= \int_{-\pi/3}^0 \frac{1}{2} (3 \cos^2 \theta + \sin^2 \theta + 2\sqrt{3} \sin \theta \cos \theta) d\theta.$$

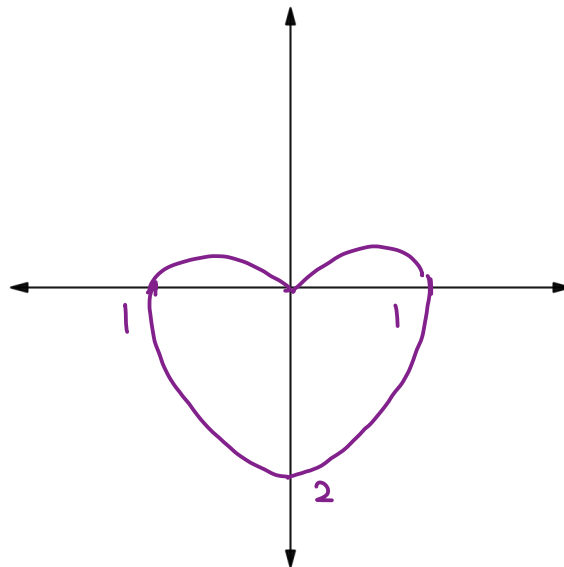
$$= \int_{-\pi/3}^0 \frac{1}{2} [1 + 2 \cos^2 \theta + \sqrt{3} \sin 2\theta] d\theta. \quad \underline{\cos 2\theta = 2 \cos^2 \theta - 1}$$

$$= \int_{-\pi/3}^0 \frac{1}{2} [1 + 1 + \cos 2\theta + \sqrt{3} \sin 2\theta] d\theta = 0 + \frac{1}{4} \sin 2\theta - \frac{\sqrt{3}}{4} \cos 2\theta \Big|_{-\pi/3}^0$$

$$= -\frac{\sqrt{3}}{4} - \left(-\frac{\pi}{3} - \frac{1}{4} \cdot \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{4} \left(-\frac{1}{2}\right)\right) = -\frac{\sqrt{3}}{4} + \frac{\pi}{3}.$$

2. (Spring 2023, #6) Consider the polar curve $r = 1 - \sin(\theta)$.

(a) Sketch the curve on the axes below.



θ	r
0	1
$\frac{\pi}{2}$	0
π	1
$\frac{3\pi}{2}$	2
2π	1

(b) Find the area enclosed by the curve.

$$A = \int_0^{2\pi} \frac{1}{2} (r(\theta))^2 d\theta .$$

$$= \int_0^{2\pi} \frac{1}{2} (1 - \sin\theta)^2 d\theta$$

$$= \int_0^{2\pi} \frac{1}{2} [1 + \sin^2\theta - 2\sin\theta] d\theta .$$

$$= \int_0^{2\pi} \frac{1}{2} + \frac{1}{2} \sin^2\theta - \sin\theta d\theta .$$

$$\cos 2\theta = 1 - 2\sin^2\theta$$

$$= \int_0^{2\pi} \frac{1}{2} + \frac{1}{2} \left(\frac{1 - \cos 2\theta}{2} \right) - \sin\theta d\theta .$$

$$= \frac{\theta}{2} + \frac{\theta}{4} - \frac{1}{8} \sin 2\theta + \cos\theta \Big|_0^{2\pi} = \frac{3}{4} \cdot 2\pi = \frac{3}{2} \pi .$$

3. (Fall 2019, #8) Consider the equation

$$r = \sin(4\theta)$$

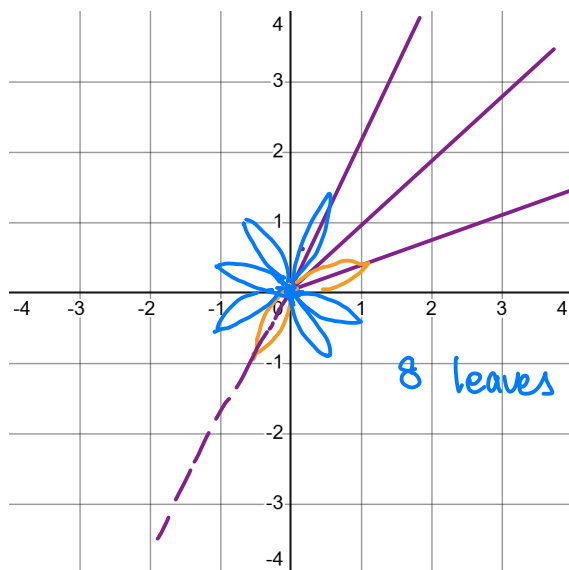
(a) In the figure below, draw the polar curve $r = \sin(4\theta)$ for all $0 \leq \theta \leq 2\pi$.

period $\frac{2\pi}{4} = \frac{\pi}{2}$.

$$4\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

$$\Rightarrow \theta = 0, \frac{\pi}{8}, \frac{\pi}{4}, \frac{3\pi}{8}, \frac{\pi}{2}$$

$$r = 0 \quad 1 \quad 0 \quad -1 \quad 0$$



(c) tangent line,

8 leaves.

(b) Find the area enclosed by the curve, in the first quadrant. (The first quadrant is the region where $x \geq 0$ and $y \geq 0$.)

$$\int_0^{\frac{\pi}{2}} \frac{1}{2} (r(\theta))^2 d\theta$$

→ This area is the orange part.

The area is the same as that in the first quadrant.

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2} (\sin(4\theta))^2 d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2} \left(\frac{1 - \cos 8\theta}{2} \right) d\theta = \frac{\pi}{8}$$

(c) Find the slope of the tangent line to the curve when $\theta = \pi/4$.

$$\text{slope} = \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta} = \frac{4 \cos 4\theta \sin \theta + \sin 4\theta \cos \theta}{4 \cos 4\theta \cos \theta - \sin 4\theta \sin \theta}$$

$$\begin{cases} y = r \sin \theta \\ x = r \cos \theta \end{cases}$$

$$r = \sin 4\theta \quad \frac{dr}{d\theta} = 4 \cos 4\theta$$

$$\text{plug in } \theta = \frac{\pi}{4} \approx \frac{4 \cdot (-1) \cdot \frac{\sqrt{2}}{2} + 0}{4 \cdot (-1) \cdot \frac{\sqrt{2}}{2} - 0} = 1$$

(d) Write an integral that computes the arc length of this curve where $0 \leq \theta \leq \pi/4$. You do NOT have to evaluate this integral.

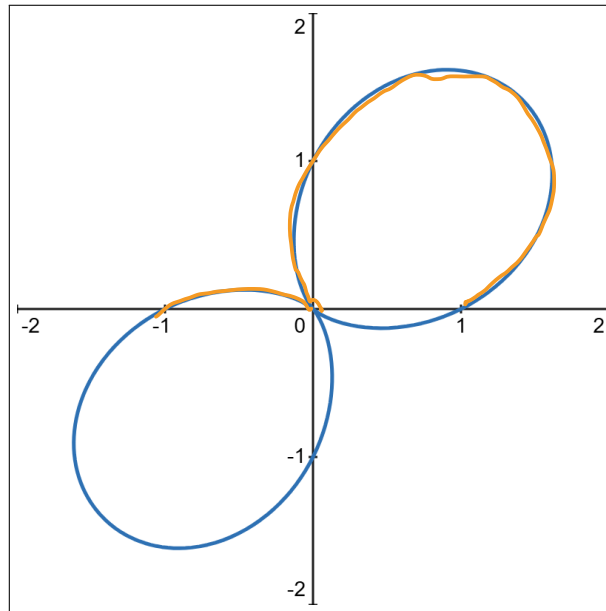
$$L = \int_a^b \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

$$= \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$L = \int_0^{\frac{\pi}{4}} \sqrt{(\sin 4\theta)^2 + (4 \cos 4\theta)^2} d\theta$$

$$\frac{dr}{d\theta} = 4 \cos 4\theta$$

4. (Spring 2018, #7) Consider the curve (pictured below) given in polar coordinates by $r = 1 + \sin(2\theta)$.



Find the area of one of the loops pictured above.

period $\frac{2\pi}{2} = \pi$,

$$\frac{1}{2} \int_0^{2\pi} \frac{1}{2} r(\theta)^2 d\theta$$

$$= \frac{1}{4} \int_0^{2\pi} (1 + \sin 2\theta)^2 d\theta .$$

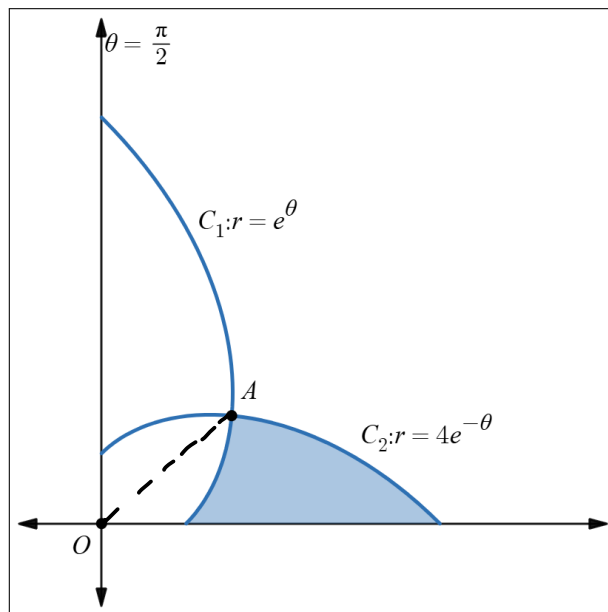
$$= \frac{1}{4} \int_0^{2\pi} (1 + \sin^2 2\theta + 2 \sin 2\theta) d\theta .$$

$$= \frac{1}{4} \int_0^{2\pi} \left(1 + \frac{1 - \cos 4\theta}{2} + 2 \sin 2\theta \right) d\theta .$$

$$= \frac{1}{4} \left[\theta + \frac{\theta}{2} - \frac{\sin 4\theta}{8} - \frac{2 \cos 2\theta}{2} \right] \Big|_0^{2\pi} .$$

$$= \frac{1}{4} \left[\frac{3}{2} \cdot 2\pi \right] = \frac{3}{4} \pi .$$

5. (Fall 2017, #6) Find the area of the shaded region below. The equations of the curves C_1 and C_2 are given in polar coordinates.



$$A: e^{\theta} = 4e^{-\theta} \Rightarrow e^{2\theta} = 4.$$

$$\Rightarrow \theta = \frac{1}{2} \ln 4 = \ln 2,$$

$$\text{Area} = \int_0^{\ln 2} \frac{1}{2} (r_2(\theta))^2 - \frac{1}{2} (r_1(\theta))^2 d\theta.$$

$$= \int_0^{\ln 2} \frac{1}{2} 16e^{-2\theta} - \frac{1}{2} e^{2\theta} d\theta.$$

$$= \int_0^{\ln 2} 8e^{-2\theta} - \frac{1}{2} e^{2\theta} d\theta \quad \begin{array}{l} \text{= } 2\frac{1}{4} \\ \text{= } (-4 \cdot \frac{1}{4} - \frac{1}{4} \cdot 4) - (-4 - \frac{1}{4}) \end{array}$$

$$= -4e^{-2\theta} - \frac{1}{4} e^{2\theta} \Big|_0^{\ln 2} = (-4e^{-2\ln 2} - \frac{1}{4} e^{2\ln 2}) - (-4 - \frac{1}{4}).$$